Assignment 4 - PIPES

//part 1- pipe

```
#include<stdio.h>
#include<sys/types.h>
#include<unistd.h>
#include<stdlib.h>
#include<sys/wait.h>

#define max 30
int main(){
    int filedes[2], ch;
    char string[max];
    char readString[max];
    pid_t pid;
    printf("enter string to be given to parent: ");
    fgets(string, max, stdin);

    if(pipe(filedes) < 0){
        printf("pipe creation error!");
        exit(0);
    }
    pid = fork();
    if(pid < 0){
        printf("Fork creation was unsuccessful!");
        exit(0);
    }
```

```
    else if(pid > 0){
        close(filedes[0]);
        write(filedes[1], string, max);
        close(filedes[1]);
        wait(NULL);
    }
    else if(pid == 0){
        close(filedes[1]);
        ch = read(filedes[0], readString, max);
        readString[ch] = '\0';
        printf("Data read by child is: %s", readString);
        close(filedes[0]);
    }
    return 0;
}
/*
OUTPUT
enter string to be given to parent: hello unnatti
Data read by child is: hello unnatti
*/
```

```
//part 2- pipe

#include<stdio.h>
#include<sys/types.h>
#include<unistd.h>
#include<stdlib.h>
#include<sys/wait.h>
#define max 30
int main(){
    int filedes1[2],filedes2[2],ch;
    char string1[max], string2[max];
    char readString[max], readString2[max];
    pid_t pid1, pid2;
    printf("enter string to be given to parent: ");
    fgets(string1, max, stdin);

    if(pipe(filedes1) < 0){
        printf("pipe creation error!");
        exit(0);
    }
    pid1 = fork();
    if(pid1 < 0){
        printf("Fork creation was unsuccessful!");
        exit(0);
    }
    else if(pid1 > 0){
        close(filedes1[0]);
        write(filedes1[1], string1, max);
```

```
    close(filedes1[1]);
    wait(NULL);
}
else if(pid1 == 0){
    close(filedes1[1]);
    ch = read(filedes1[0], readString, max);
    readString[ch] = '\0';
    printf("Data read by child of pipe 1 is: %s", readString);
    close(filedes1[0]);
}
pid2 = fork();
printf("\nenter string to be given to child: ");
fgets(string2, max, stdin);
if(pipe(filedes2) < 0){
    printf("pipe creation error!");
    exit(0);
}
if(pid2 < 0){
    printf("Fork creation was unsuccessful!");
    exit(0);
}
else if(pid2 == 0 ){
    close(filedes2[0]);
    write(filedes2[1], string2, max);
    close(filedes2[1]);
    wait(NULL);
}
```

```
    else if(pid2 > 0){
        close(filedes2[1]);
        ch = read(filedes2[0], readString2, max);
        readString[ch] = '\0';
        printf("Data read by parent of pipe 2 is: %s", readString2);
        close(filedes2[0]);
    }
        return 0;


}
```